Central and Peripheral Routes to Collective Belief: An Agent-Based Model of Mixed-Processing Populations

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Abstract

People differ in how they process arguments. Some evaluate the evidential content of a message (the central route of the Elaboration Likelihood Model). Others track only the opinions around them (the peripheral route). What happens when both kinds of processor share one communication network is poorly understood, because most models of belief formation assume a single updating rule. We built an agent-based model in which Bayesian NormAN agents (central route) and DeGroot-averaging contagion agents (peripheral route) share one Bayesian world model and one network. We crossed the proportion of NormAN agents (0 to 1) with argument-selection strategy (random vs. impact) and communication scope (pairwise vs. broadcast) across 4,400 runs, and ran a 17,600-run extension that varied network structure. We tested six a priori hypotheses. Group error fell strictly with NormAN proportion (from .314 to .058), but non-linearly: gains were concentrated between 10% and 40% NormAN, which points to a critical mass rather than an immediate benefit. Impact-based selection gave no advantage over random sharing. Broadcast communication improved accuracy (.122 vs. .192) and reduced polarisation. Placing peripheral processors at network hubs degraded accuracy far more than placing them at the periphery. Peripheral processors benefited disproportionately from mixing. An exploratory analysis found that mixing itself produced polarisation, which peaked in mixed populations even though the model contains no negative influence, homophily, or biased assimilation. Collective epistemic performance depended on population composition, communication reach, and network structure rather than on individual rationality.

Keywords: Elaboration Likelihood Model; agent-based modelling; opinion dynamics; Bayesian argumentation; social epistemology; polarisation; NormAN

1 Introduction

Some people engage carefully with the content of the arguments they meet. Others rely on simpler cues, such as how popular a claim is or how credible its source seems. The difference is the one between reading a research paper and retweeting its headline. It has been central to the psychology of persuasion since Petty and Cacioppo [1986] introduced the Elaboration Likelihood Model (ELM). Despite decades of work on persuasion and on opinion dynamics, we know little about what happens at the level of a population when both kinds of processor share one communication network. Most computational models of belief formation give every agent the same updating rule, whether Bayesian or heuristic. In this paper we combine agent-based modelling with the ELM's dual-process framework to study how populations that contain both careful reasoners and heuristic agents arrive at collective beliefs.

1.1 The Elaboration Likelihood Model

The ELM distinguishes two routes by which a persuasive message can change an attitude [Petty and Cacioppo, 1986]. On the central route, the recipient evaluates the message with effort: they attend to its logical structure and evidential content and weigh its merits against what they already know. On the peripheral route, the recipient relies on surface cues, such as the attractiveness of the source, the length of the message, the number of arguments, or how other people react. The probability that a person takes the central route, their elaboration likelihood, depends on motivation (the personal relevance of the topic, dispositional need for cognition) and on ability (cognitive load, time pressure, domain expertise). A key prediction is that attitudes formed via the central route are more durable, more resistant to counter-persuasion, and more predictive of behaviour [Petty and Wegener, 1999].

The ELM is one of the most cited theories in social psychology, but it has remained a theory of how individuals process messages. It has not been used to define agent types in a computational model of population-level belief dynamics. A partial exception is Mosler et al. [2001], who used simulation to formalise the ELM for a single person, modelling how processing intensity shifts with argument quality and peripheral cues. They did not create distinct agent types or place them in a social network. Their aim was to formalise the individual-level theory, not to scale it up. We take that step here.

1.2 Opinion dynamics and social epistemology

Two modelling traditions are relevant. The first, which starts with DeGroot [1974], treats belief formation as weighted averaging. At each step every agent takes a convex combination of its own opinion and those of its network neighbours. DeGroot dynamics are tractable and converge in well-understood ways. Golub and Jackson [2010] proved that DeGroot agents in large networks converge to the truth if and only if no single agent is too influential, that is, if the largest influence weight vanishes as the network grows. The model's simplicity is both a strength and a weakness. Its agents have no access to evidence, no capacity for reasoning, and no representation of argument structure. In ELM terms they are pure peripheral-route processors.

The second tradition gives agents Bayesian rationality. In models of Bayesian social learning [for a review, see Golub and Sadler, 2016], agents observe private signals, form posteriors by Bayes' rule, and may pass beliefs or evidence to their neighbours. These agents are ideal central-route processors. Bayesian rationality does not guarantee good collective outcomes, however. Madsen et al. [2018] showed that even error-free Bayesian agents form persistent echo chambers, and that larger networks can make this worse rather than better. Individual rationality is neither necessary nor sufficient for collective epistemic success. What matters is how agent properties interact with communication structure.

Computational social epistemology has developed this point at length. Zollman [2007] showed that less connected networks of inquirers can do better because they preserve diversity of inquiry, a result now called the Zollman effect. Rosenstock et al. [2017] showed that the effect is fragile and can reverse under modest parameter changes. Conformity pressures can produce persistent disagreement even among agents who gather evidence [Weatherall and O'Connor, 2020], and agents who discount evidence from people with dissimilar beliefs produce stable scientific polarisation [O'Connor and Weatherall, 2018]. More broadly, Zollman [2010] argued for the epistemic benefit of transient diversity, and Hong and Page [2004] proved that cognitively diverse groups can outperform groups of individually high-ability problem solvers.

This literature shows that the composition of a population matters for collective epistemics,

not only the rationality of its members or the topology of its network. Yet the heterogeneity studied so far usually stays within one modelling paradigm. DeGroot-style models vary weights, stubbornness, or confidence thresholds [Ghaderi and Srikant, 2014, Mirtabatabaei and Bullo, 2012]. Bayesian models vary communication strategies [Schöppl and Hahn, 2024]. To our knowledge no published model has placed Bayesian reasoners and DeGroot-style averagers in the same network to study how they interact. That is the heterogeneity we introduce.

1.3 The NormAN framework

Our model builds on NormAN (Normative Argument Exchange across Networks; Assaad et al., 2023, 2025). In NormAN, agents sit in a social network and exchange arguments about a shared hypothesis. Each agent holds a Bayesian network as its world model and updates its belief in the hypothesis by exact Bayesian inference as it gathers evidence. Arguments are pieces of evidence: observations of variables in the Bayesian network that bear, directly or indirectly, on the hypothesis. Agents share arguments with their neighbours by one of two strategies. Under random sharing, an agent picks an argument uniformly at random from its evidence. Under impact sharing, it picks the argument that most strongly supports its current belief.

Assaad et al. [2023] showed that impact sharing produces a hidden-profile effect. Agents preferentially circulate belief-consistent evidence, disconfirming arguments diffuse less, and collective accuracy suffers. This mirrors a well-known phenomenon in group decision making. In a meta-analysis of hidden-profile studies, Lu et al. [2012] found that groups discuss shared information much more than unique information, and that hidden-profile groups are about eight times less likely than fully informed groups to find the correct solution. Del Vicario et al. [2017] showed in a model that confirmation bias and social influence together produce polarisation.

NormAN is modular: the world model, agent architecture, communication protocol, and network can be varied independently. Schöppl and Hahn [2024] used this to study heterogeneity in communication strategy. A proportion of their agents were "agreeable": they self-censored by matching the polarity of their neighbours' last argument, sharing only evidence consistent with the local majority. Small proportions of agreeable agents degraded accuracy, and the relationship between their proportion and population outcomes was non-linear. Heterogeneity among NormAN agents can thus produce effects that cannot be read off individual behaviour. In that model, though, every agent was a Bayesian reasoner. Agents differed in which arguments they shared, not in whether they engaged with argument content at all.

NormAN belongs to the wider family of argumentative agent-based models (AABMs), recently surveyed by Dupuis de Tarlé et al. [2025]. Alternatives include the abstract argumentation model of Borg et al. [2018], which uses formal argumentation theory rather than Bayesian networks to represent scientific debate. What sets NormAN apart is the Bayesian network as shared world model. It gives a natural measure of accuracy (the distance between an agent's belief and the correct posterior given the available evidence) and a principled account of how pieces of evidence combine. This makes it a good basis for asking what happens when agents with different processing capacities are mixed.

1.4 The present study

We extend NormAN with a second agent type, the contagion agent, which operationalises peripheral-route processing. NormAN agents evaluate argument structure by Bayesian inference. Contagion agents average the opinions of their neighbours in the manner of DeGroot [1974]. They hold no Bayesian network, perform no inference, and do not gather evidence. When they receive an

argument they cannot evaluate its content. Instead they forward it if its direction agrees with their current belief. The contagion agent is meant to capture a social media user who shares congenial content without engaging with the reasoning behind it.

We mix the two types in varying proportions in a Watts–Strogatz small-world network [Watts and Strogatz, 1998] under four communication conditions, formed by crossing argument selection (random vs. impact) with communication scope (pairwise vs. broadcast). Pairwise communication models one-to-one interaction. Broadcast communication, in which an agent sends to all of its neighbours at once, is an analogue of the one-to-many architecture of social media. The proportion of NormAN agents runs from 0% (all contagion) to 100% (all NormAN) in steps of 10%. A second experiment uses Barabási–Albert scale-free networks [Barabási and Albert, 1999] to ask whether it matters which type occupies the hubs.

The question has practical as well as theoretical interest. Bak-Coleman et al. [2021] argued that the study of collective behaviour should become a "crisis discipline", because new communication technologies have changed information flows in ways we do not understand. On social media platforms a minority of users produce arguments and a majority consume and redistribute them with varying degrees of critical engagement. Knowing when such mixed populations reach accurate beliefs bears on how communication environments are designed.

We tested six a priori hypotheses:

H1 *Composition.* Because NormAN agents track evidence and contagion agents do not, group error decreases monotonically as the proportion of NormAN agents rises.

H2 *Disproportionate early gain.* Because contagion agents forward arguments they agree with, a small number of NormAN agents produces a disproportionate gain in accuracy, with diminishing returns thereafter.

H3 *Argument selection.* Impact-based selection produces lower final error than random selection, particularly at moderate NormAN proportions.

H4 *Communication scope.* Broadcast communication reduces polarisation faster than pairwise communication (H4a), but does not reliably reduce error (H4b), because faster consensus may lock in premature conclusions.

H5 *Hub occupation.* In scale-free networks, contagion agents at hub positions harm group accuracy more than contagion agents at peripheral positions.

H6 *Asymmetric benefit.* Contagion agents are more accurate when mixed with NormAN agents, and this gain is larger than the corresponding gain for NormAN agents.

We also report one exploratory analysis, not among the hypotheses, of whether mixing itself affects polarisation.

2 Model and Methods

The model contains two agent types, one for each route of the ELM. Every agent i holds a degree of belief $b_i(t) \in [0, 1]$ in a target proposition H at each time step t. The types differ in how they update this belief in response to what their neighbours send them. A summary following the ODD protocol is given in Appendix B.

2.1 NormAN agents

NormAN agents follow the NormAN framework [Assaad et al., 2023]. Each holds a local copy of a Bayesian network and a set $E_i(t)$ of evidence nodes it has observed by time t. Its belief in H is the posterior

$$b_i(t) = P\big(H \mid E_i(t)\big), \tag{1}$$

computed by exact inference (variable elimination) with the `pgmpy` library [Ankan and Panda, 2015].

When a NormAN agent communicates, it selects one argument (evidence node) from its evidence set and sends it, together with its current belief $b_i(t)$, to each of its targets for that step: one randomly chosen neighbour under pairwise scope, or all neighbours under broadcast scope. What it sends does not depend on the recipient's type; the recipient decides how to use the message (Section 2.4). We compare two selection strategies:

- **Random selection.** The agent picks an argument uniformly at random from its evidence set.
- **Impact selection.** The agent picks the argument whose removal would most change its own posterior, that is, the piece of evidence currently contributing most to its belief:

$$e^* = \arg\max_{e \in E_i(t)} \big|P\big(H \mid E_i(t)\big) - P\big(H \mid E_i(t) \setminus \{e\}\big)\big|. \tag{2}$$

  The rule is sender-centric. It is evaluated over the sender's own evidence and uses no model of the recipient's beliefs or evidence.

On receiving an argument e_j from neighbour j, a NormAN agent adds it to its evidence, $E_i(t+1) = E_i(t) \cup \{e_j\}$, and recomputes Equation 1. The update is fully Bayesian: the new evidence is a hard observation in the shared causal model.

NormAN agents also inquire. At each step, alongside communication, a NormAN agent observes a new evidence node with probability $p_{\text{inq}} = 0.05$, the base rate used by Assaad et al. [2023].

2.2 Contagion agents

Contagion agents model the peripheral route. They hold no Bayesian network and do not evaluate argument structure. Each holds a scalar belief $b_i(t) \in [0, 1]$ and updates it by plain DeGroot averaging [DeGroot, 1974, French, 1956] over its neighbours $N(i)$:

$$b_i(t+1) = (1-\alpha)\, b_i(t) + \alpha \, \frac{1}{|N(i)|} \sum_{j \in N(i)} b_j(t), \tag{3}$$

where $\alpha \in (0, 1)$ is a susceptibility parameter that sets the weight on neighbours' beliefs relative to the agent's own. We set $\alpha = 0.3$ on the basis of a preliminary sensitivity analysis. Contagion agents do not inquire, so they cannot discover evidence, and they cannot evaluate arguments. Their beliefs change only through social influence. This asymmetry is deliberate: it reflects the ELM's description of peripheral processing as low-effort, cue-driven, and dependent on social context.

Contagion agents do take part in argument transmission. When a contagion agent receives an argument from a NormAN neighbour, it forwards it to its own neighbours only if the argument's direction agrees with its current lean:

$$\text{Forward}(e, i) = \begin{cases} \text{true} & \text{if } \operatorname{sgn}(\delta_e) = \operatorname{sgn}\big(b_i(t) - 0.5\big), \\ \text{false} & \text{otherwise,} \end{cases} \tag{4}$$

where $\delta_e = P(H \mid \{e\}) - P(H)$ is the direction of support that e gives H in the shared Bayesian network (positive if e supports H). An argument that supports H is passed on by an agent with $b_i > 0.5$, and an argument against H by an agent with $b_i < 0.5$. This "retweet bot" rule is a minimal confirmation-bias heuristic. It lets contagion agents act as selective conduits for arguments without evaluating them.

2.3 World model

All NormAN agents share the "Vole" Bayesian network [Fenton and Neil, 2018], an artificial network with 22 nodes and 25 arcs, of which 7 nodes serve as evidence. It has been the main benchmark in earlier NormAN work [Assaad et al., 2023, Schöppl and Hahn, 2024]. In each run H is set to true, and the values of the evidence nodes are drawn by conditioning the network on that value. The network suits our purpose for three reasons. It is small enough for exact inference, so no approximation noise enters. Its 7 evidence nodes give several evidential pathways to H, so different subsets of evidence can support or undermine it. And using the same benchmark as Schöppl and Hahn [2024] allows direct comparison with their results on self-censoring. NormAN agents share the network's structure and conditional probability tables and differ only in which evidence they have absorbed. Contagion agents have no copy of the network, but the network still fixes the valence δ_e of the arguments they forward.

2.4 Cross-type communication

The two types represent belief in different formats (evidence sets vs. scalar credences), so we must specify how they exchange information. The protocol is asymmetric, reflecting the asymmetry between the two routes.

NormAN to contagion. The NormAN agent sends an argument as usual. The contagion agent does not evaluate its content. It (a) includes the sender's belief $b_j(t)$ in its averaging update, treating the NormAN agent like any other neighbour, and (b) applies the forwarding rule (Equation 4) to decide whether to pass the argument on. Contagion agents are therefore influenced by what NormAN agents believe, but not by why they believe it.

Contagion to NormAN. The contagion agent sends either an argument it is forwarding, which the NormAN agent incorporates as genuine evidence, or its current belief $b_i(t)$, which the NormAN agent discards, since a credence without evidential grounding carries no inferential weight in a Bayesian framework. Arguments keep their evidential identity whoever relays them, so NormAN agents can benefit from evidence relayed through contagion agents but are not swayed by their unevidenced opinions. Contagion agents thus act as selective argument relays, much like social media users who share content that resonates with their views without evaluating the reasoning behind it.

2.5 Network topology

Agents occupy the nodes of an undirected network $G = (V, E)$ whose edges are communication channels. The primary topology is a Watts–Strogatz small-world network [Watts and Strogatz, 1998], with high clustering and short average path lengths, as in many real social networks. It has $N = 20$ agents, each initially linked to its $K = 2$ nearest neighbours on each side (degree $2K = 4$),

with each edge rewired to a random node with probability $p_{\text{rw}} = 0.2$. These values follow Assaad et al. [2023] and Schöppl and Hahn [2024].

In small-world networks, contagion agents at peripheral positions can become dead ends for arguments: they receive them but have few neighbours to forward them to. Scale-free networks instead have a few highly connected hubs that could act as conduits. As an extension we therefore also use Barabási–Albert preferential-attachment networks [Barabási and Albert, 1999], generated with NetworkX's `barabasi_albert_graph` with $n = 20$ and $m = 2$ (each new node attaches to two existing nodes). This gives a mean degree of about 4, comparable to the small-world condition, but with a heavy-tailed degree distribution. A hub-occupation factor assigns either NormAN or contagion agents preferentially to the highest-degree nodes.

2.6 Communication protocols

Communication follows one of four protocols in a 2×2 design. Argument selection (random or impact; Section 2.1) governs which argument a NormAN agent sends. Contagion agents do not select arguments; they forward received arguments by the valence rule and send their current belief. Communication scope governs reach. Under *pairwise* scope, each agent communicates with one randomly chosen neighbour per step. Under *broadcast* scope, each agent sends to all of its neighbours at once. The four combinations (random–pairwise, random–broadcast, impact–pairwise, impact–broadcast) separate the effect of argument choice from that of communication reach.

2.7 Experimental design

The primary design is a $2 \times 2 \times 11$ factorial (argument selection $\times$ communication scope $\times$ NormAN proportion) on Watts–Strogatz networks, giving 44 conditions (Table 1). Each condition was replicated 100 times with independently generated networks and initial evidence, for 4,400 runs. The extension adds network topology and hub occupation as factors, giving $2 \times 2 \times 11 \times 2 \times 2 = 176$ conditions and 17,600 runs. In the primary design, agent types are assigned to network positions uniformly at random. In the hub-occupation extension, the highest-degree nodes are preferentially assigned to one type.

Table 1: Experimental factors and levels.

Factor	Levels	Values
Argument selection	2	Random, impact
Communication scope	2	Pairwise, broadcast
NormAN proportion	11	$0.0, 0.1, \ldots, 1.0$
Extension only		
Network topology	2	Watts–Strogatz, Barabási–Albert
Hub occupation	2	NormAN at hubs, contagion at hubs

2.8 Initialisation and schedule

At $t = 0$ a network is generated with the parameters above and agents are assigned types according to the condition's NormAN proportion. Following Schöppl and Hahn [2024], each NormAN agent starts with exactly one piece of evidence ($k_0 = 1$), drawn at random from the evidence nodes, and its initial belief is the posterior given that evidence. NormAN agents communicate once per step and always communicate, whatever their confidence (conviction threshold 0). Each contagion

agent's initial belief is drawn from Beta$(2, 2)$, a mildly dispersed distribution centred on 0.5 that represents initial uncertainty. Table 2 lists all fixed parameters.

Each run lasts $T = 25$ steps, following Schöppl and Hahn [2024]. Algorithm 1 gives the schedule of one step. Activation order is shuffled each step to prevent order effects. Belief updates are applied synchronously after all communication in a step, so that early-activated agents cannot trigger within-step cascades.

Algorithm 1 One simulation step t.

Shuffle agent activation order uniformly at random
for each agent i in shuffled order **do** ▷ communication phase
  if i is a NormAN agent **then**
    With probability p_{inq}, observe a new evidence node
    Select argument e^* by the condition's selection strategy
    Send e^* and current belief $b_i(t)$ to communication targets
  else ▷ contagion agent
    Send current belief $b_i(t)$ to communication targets
    Forward queued arguments according to Equation 4
  end if
end for
for each agent i **do** ▷ synchronous belief update
  if i is a NormAN agent **then**
    Add received arguments to E_i and recompute $b_i(t+1)$ (Equation 1)
  else
    Update $b_i(t+1)$ by Equation 3
  end if
end for
Record all beliefs and network state for step $t+1$

Table 2: Fixed model parameters.

Component	Parameter	Value
Population	Number of agents N	20
Watts–Strogatz	Neighbours per side K; rewiring probability p_{rw}	2; 0.2
Barabási–Albert	Edges per new node m	2
World model	Vole network: nodes / arcs / evidence nodes	22 / 25 / 7
NormAN agents	Inquiry probability p_{inq}	0.05
	Initial evidence k_0	1
	Messages per step; conviction threshold	1; 0
Contagion agents	Susceptibility α	0.3
	Initial belief	Beta$(2, 2)$
	Forwarding threshold on b_i	0.5
Schedule	Steps T	25
	Convergence tolerance ε	0.01
Design	Replications per condition	100

2.9 Outcome measures

Following Angere and Olsson [2017] and Schöppl and Hahn [2024], *group error* is the root mean squared distance between agents' beliefs and the optimal posterior $p_{\text{opt}}(H)$, the posterior obtained by conditioning on all evidence available in the run:

$$\text{Error}(t) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(b_i(t) - p_{\text{opt}}(H)\big)^2}. \tag{5}$$

Polarisation is the root mean squared distance between agents' beliefs and the population mean:

$$\text{Pol}(t) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(b_i(t) - \bar{b}(t)\big)^2}, \qquad \bar{b}(t) = \frac{1}{n}\sum_{i=1}^{n} b_i(t). \tag{6}$$

Higher values mean more disagreement. The two measures are complementary. A population can agree on the wrong answer (low polarisation, high error) or disagree around an accurate mean (high polarisation, low error). We also recorded *convergence*, the first step at which $\text{Pol}(t) < \varepsilon$, with runs that never reach it within T steps counted as non-convergent; and *type-specific error*, Equation 5 computed separately over NormAN and over contagion agents, to see whether one type benefits more from the other's presence.

2.10 Analysis

Unless stated otherwise, outcomes are measured at the final step ($T = 25$). Group error and polarisation residuals departed mildly from normality, but with 100 replications per cell the sampling distributions of condition means are close to normal (details in Appendix A). We therefore used parametric tests and checked every key pairwise comparison with a Mann–Whitney U test. The non-parametric result agreed with the parametric one in every case. With this many replications, trivial differences reach significance, so we report effect sizes (Cohen's d, partial η^2) throughout and base verdicts on magnitude as well as significance (two-tailed, at the .05 level). We did not correct for multiple comparisons, because no verdict below would change under any standard correction.

2.11 Implementation

The simulation is written in Python 3.11. Graphs use NetworkX [Hagberg et al., 2008], inference uses `pgmpy` [Ankan and Panda, 2015], and analysis uses NumPy, pandas, and Matplotlib. The original NormAN framework and its self-censoring extension are implemented in NetLogo [Wilensky, 1999], with inference through R and the `bnlearn` [Scutari, 2010] and `gRain` [Højsgaard, 2012] packages. We moved to Python for two reasons. A DeGroot agent type is a fundamentally different kind of agent and is more naturally written as a Python class than as NetLogo breed-level procedures. And `pgmpy` gives native exact inference without cross-language calls, which simplifies reproduction. We did not use a general Python ABM framework such as Mesa. The computational core is the inference and the cross-type communication, which must be written directly in any case, and a framework-free loop gives full control over the update order.

3 Results

Table 3 summarises the outcome for each hypothesis. The sections below give the details.

Table 3: Summary of hypothesis tests. Means are final-step values. W: Watts–Strogatz primary design (4,400 runs). BA: Barabási–Albert conditions of the extension, mixed populations only.

Hypothesis	Key result	Verdict
H1: error falls with NormAN share	$M = .314 \rightarrow .058$, strictly decreasing; $R^2 = .41$	Supported
H2: small NormAN share gives disproportionate gain	Largest gains between 10% and 40%; first 10% gives .017	Partly supported (critical mass)
H3: impact beats random	Impact slightly worse ($M = .163$ vs. .151, $d = 0.09$); null at .3–.6	Not supported
H4a: broadcast lowers polarisation	$M = .084$ vs. .128, $d = -1.18$	Supported
H4b: broadcast does not lower error	$M = .122$ vs. .192, $d = -0.55$	Contradicted
H5: contagion at hubs is worse (BA)	$M = .209$ vs. .160, $d = 0.38$; halved in W	Supported
H6: contagion agents gain more from mixing	Slope $-.272$ vs. $-.236$; gap narrows	Supported

3.1 H1: Group accuracy rises with the share of NormAN agents

NormAN-only populations converge on well-calibrated beliefs [Assaad et al., 2023], while pure DeGroot averaging has no way to track evidence. More truth-trackers should therefore lower group error. They did. Mean final error fell from $M = .314$ ($SD = .118$) in pure-contagion populations to $M = .128$ ($SD = .102$) at the 50% midpoint and $M = .058$ ($SD = .060$) in pure-NormAN populations, decreasing strictly across all eleven proportions (Figure 1, left). Run-level distributions shifted down in the same way, and their spread contracted as the NormAN proportion rose (Figure 1, right). The rank-order association between NormAN proportion and final error was strong and negative, Spearman's $\rho(4398) = -.65$, $p < .001$. In a linear regression, each 10-percentage-point increase in NormAN proportion reduced final error by .027 on average, $b = -.270$, $t(4398) = -55.68$, $p < .001$, $R^2 = .41$. H1 was supported.

3.2 H2: The effect of mixing has a threshold

Contagion agents forward arguments they agree with, so even a few NormAN agents' arguments might travel far beyond their own neighbourhoods. H2 accordingly predicted a disproportionate gain from a small number of NormAN agents, with diminishing returns thereafter.

The relationship was indeed non-linear. Gains were concentrated between 10% and 40% NormAN and flattened at high proportions (Figure 2, which plots the first differences of the means in Figure 1). Adding a quadratic term improved fit over the linear model, $F(1, 4397) = 222.77$, $p < .001$, $\Delta R^2 = .028$ (quadratic $b = .253$, $t = 14.93$, $p < .001$).

Where the steepest gains fell, however, differed from the prediction. The first 10% of NormAN agents reduced error only modestly (.017). The largest gains came between 10% and 40% (.053, .051, and .041 per step), and gains fell below .010 per step beyond 80%. This is a critical-mass pattern: accuracy gains accelerate once NormAN agents make up roughly a fifth of the population, rather than arriving with the first few careful reasoners. H2 was supported in a modified form.

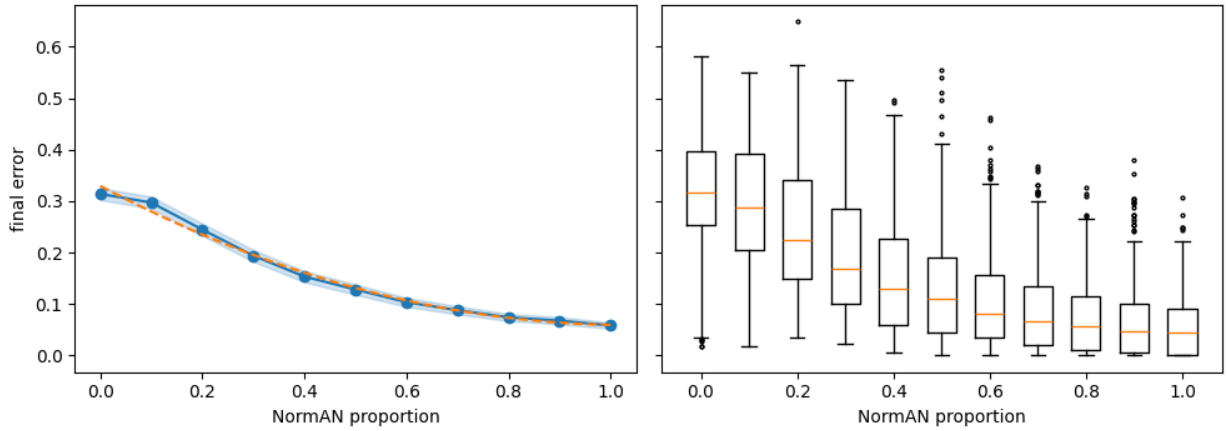


Figure 1: Final group error against the proportion of NormAN agents (primary design, 4,400 runs). Left: condition means (points and solid line, with shaded uncertainty band) and a fitted trend (dashed). Error falls at every step, from $M = .314$ in pure-contagion populations to $M = .058$ in pure-NormAN populations. Right: run-level distributions at each proportion (box: interquartile range; centre line: median; whiskers: 1.5×IQR; points: outliers). The distribution shifts down and narrows as the NormAN proportion rises.

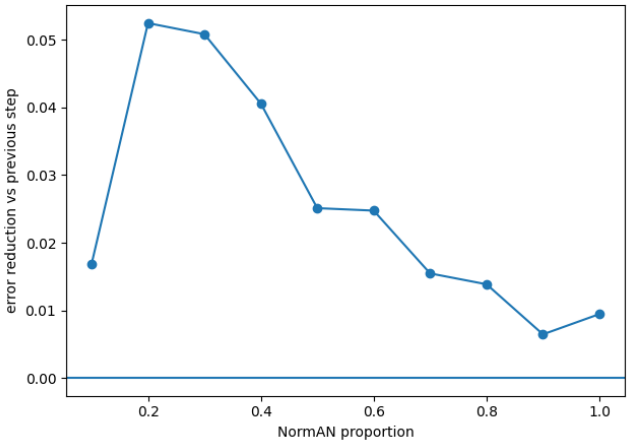


Figure 2: Marginal reduction in group error for each 10-percentage-point increase in NormAN proportion, computed as successive differences of the condition means in Figure 1. The first increment gives a small gain (about .017). Gains peak between 10% and 40% NormAN (about .053, .051, and .041 per step) and fall below .010 per step beyond 80%. Accuracy gains accelerate once careful reasoners exceed roughly a fifth of the population.

3.3 H3: Impact-based selection does not outperform random sharing

H3 predicted that impact selection, in which senders transmit the argument that contributes most to their own current belief, would produce lower final error than random selection, particularly at moderate NormAN proportions. In the $2 \times 2 \times 11$ factorial ANOVA on final error, impact selection produced slightly *higher* error ($M = .163$) than random selection ($M = .151$), $F(1, 4356) = 19.20$, $p < .001$, partial $\eta^2 = .004$, $d = 0.09$. The main effect was significant but negligible, and in the opposite direction to the prediction.

At the moderate proportions where an advantage was predicted (.3 to .6), the two strategies could not be distinguished (Figure 3), $t(1598) = -0.14$, $p = .885$, $d = -0.01$, a null confirmed by a Mann–Whitney test ($p = .18$). A strategy $\times$ proportion interaction, $F(10, 4356) = 6.77$, $p < .001$, partial $\eta^2 = .015$, reflected a small advantage for random selection that emerged only at high NormAN proportions. There was also a strategy $\times$ scope interaction, $F(1, 4356) = 49.03$, $p < .001$, partial $\eta^2 = .011$. Impact-based selection does not improve accuracy at population scale. H3 was not supported.

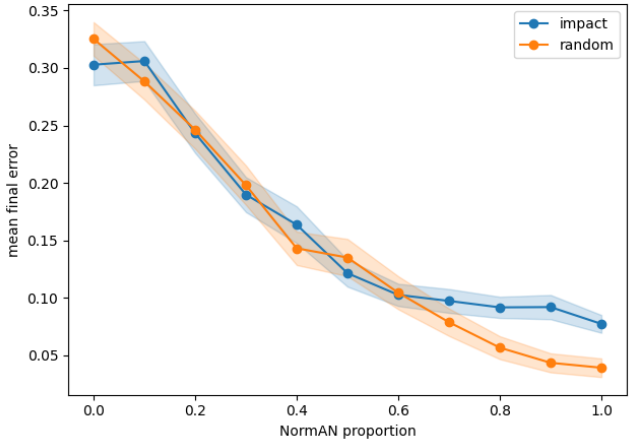


Figure 3: Mean final group error against NormAN proportion for impact-based and random argument selection (primary design, 4,400 runs; shaded bands show uncertainty around the condition means). The two strategies track each other closely at low and moderate proportions. Random selection pulls ahead only at high NormAN proportions. Averaged over all conditions, impact selection gave marginally higher error ($M = .163$ vs. $M = .151$), the opposite of the predicted direction.

3.4 H4: Broadcast communication lowers both polarisation and error

H4 made two predictions: broadcast communication would reduce polarisation faster than pairwise communication (H4a) but would not reliably reduce error (H4b).

H4a was strongly supported. Averaged over all 26 recorded time points of each run, broadcast runs were much less polarised ($M = .084$) than pairwise runs ($M = .128$), $t(4398) = -39.26$, $p < .001$, $d = -1.18$, the largest effect in the study. The trajectories separated from the first steps (Figure 4, left). Final polarisation showed the same pattern ($M = .047$ vs. $M = .092$), $t(4398) = -31.04$, $p < .001$, $d = -0.94$. Broadcast runs were also far more likely to reach the convergence criterion (polarisation below .01) within 25 steps: 25% of broadcast runs against 1.4% of pairwise runs, $\chi^2(1) = 533.37$, $p < .001$.

H4b was contradicted. Broadcast communication did not leave accuracy unchanged; it substantially reduced final error ($M = .122$) relative to pairwise communication ($M = .192$), at every

mixing ratio (Figure 4, right), $t(4398) = -18.20$, $p < .001$, $d = -0.55$. After NormAN proportion, scope was the largest effect in the primary factorial (partial $\eta^2 = .136$). We found no sign of the hypothesised premature-consensus penalty within the 25-step horizon: faster convergence went with greater accuracy, not less.

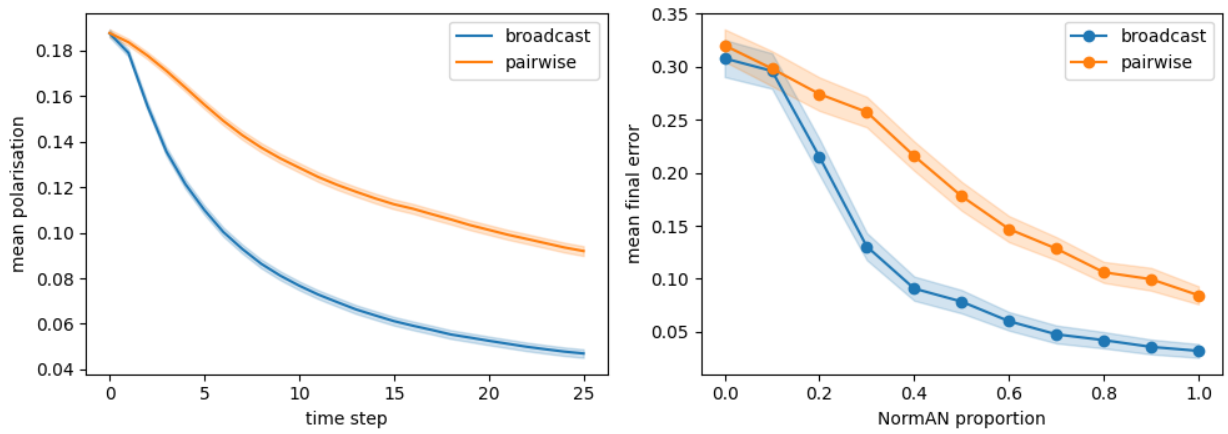


Figure 4: Effect of communication scope (primary design, 4,400 runs; shaded bands show uncertainty around the means). Left: mean polarisation over the 26 recorded time points. Broadcast runs separate from pairwise runs within the first few steps and stay much less polarised (time-averaged $M = .084$ vs. $M = .128$). Right: mean final group error against NormAN proportion. Broadcast gives lower error at every mixing ratio ($M = .122$ vs. $M = .192$), the largest effect in the primary design after population composition.

3.5 H5: Who sits at the hubs matters

H5 predicted that in scale-free networks, contagion agents at hub positions would harm group accuracy more than contagion agents at the periphery. In the Barabási–Albert conditions of the extension (mixed populations only, proportions .1 to .9), runs with contagion agents at the highest-degree nodes had higher final error ($M = .209$) than runs with NormAN agents there ($M = .160$), $t(7198) = 16.13$, $p < .001$, $d = 0.38$ (Figure 5, left). The main effect of hub occupation was $F(1, 7182) = 335.04$, $p < .001$, partial $\eta^2 = .045$. A hub $\times$ proportion interaction, $F(8, 7182) = 15.47$, $p < .001$, partial $\eta^2 = .017$, reflected a gap that was widest at low-to-moderate NormAN proportions and closed as either type came to dominate the population.

If the effect works through hub centrality, it should be smaller where hubs barely exist. In the Watts–Strogatz conditions, where degree varies little, the same comparison gave an effect half as large ($M = .163$ vs. $M = .139$, $d = 0.19$; Figure 5, right). Hub occupation matters most where there are hubs to occupy. H5 was supported.

3.6 H6: Contagion agents gain more from mixing

H6 predicted that contagion agents would be more accurate when mixed with NormAN agents, and that this gain would exceed the gain for NormAN agents. Across all mixed runs, contagion agents were less accurate than NormAN agents in the same run (mean within-run gap = .032), $t(3599) = 20.19$, $p < .001$. But the gap narrowed as NormAN proportion rose, $b = -.035$, $t(3598) = -5.78$, $p < .001$, which shows as converging error curves in Figure 6. Correspondingly, contagion agents'

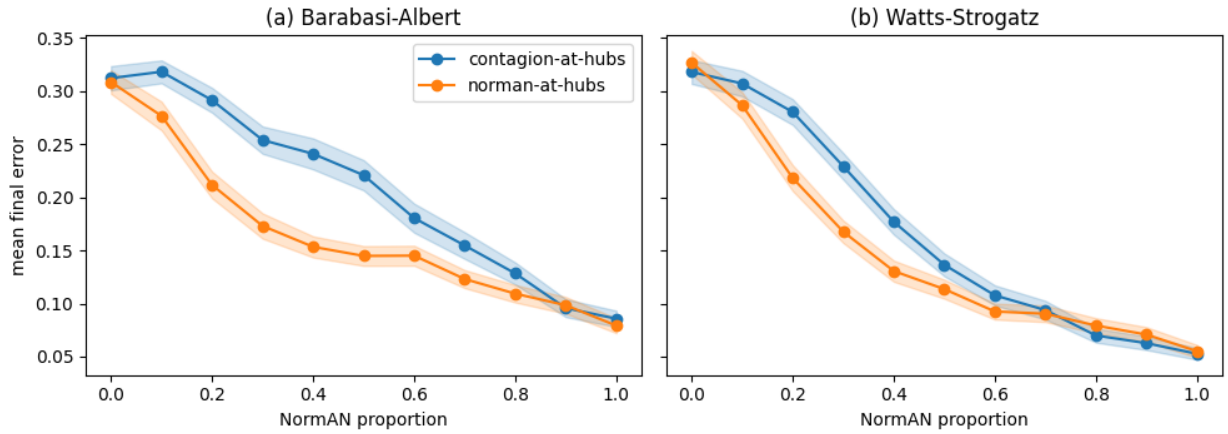


Figure 5: Mean final group error against NormAN proportion when the highest-degree nodes are seeded with contagion agents (blue) or with NormAN agents (orange), from the extension design; shaded bands show uncertainty around the means. (a) Barabási–Albert scale-free networks. Over mixed populations (.1 to .9), contagion agents at the hubs raise error ($M = .209$) relative to NormAN agents at the hubs ($M = .160$). (b) Watts–Strogatz small-world networks, where degree varies little. The same contrast is about half as large ($M = .163$ vs. $M = .139$), which ties the effect to hub centrality rather than to the seeding procedure.

error fell more steeply with NormAN proportion ($b = -.272$) than NormAN agents' own error did ($b = -.236$). Contagion agents were pulled progressively toward the truth by the growing share of evidence-based neighbours, but never fully matched the accuracy of agents who evaluate arguments directly. H6 was supported.

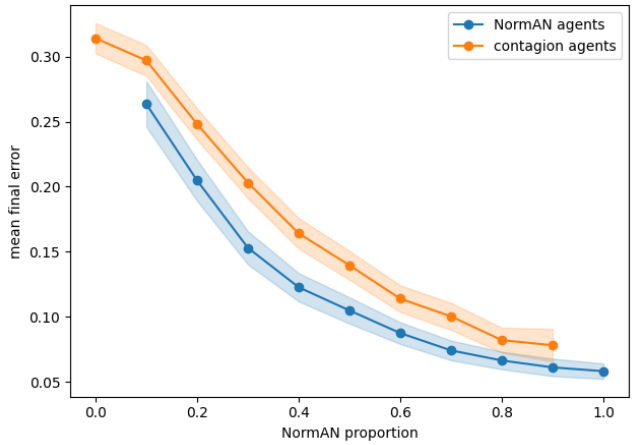


Figure 6: Mean final error of NormAN agents and contagion agents, computed separately within the same runs, against NormAN proportion (primary design; shaded bands show uncertainty around the means). Contagion agents are less accurate at every mixing ratio, but their error falls more steeply as the NormAN share grows, so the curves converge. A residual gap persists throughout. Each type's curve is undefined at the endpoint where that type is absent.

3.7 Exploratory analysis: Mixing itself produces polarisation

Beyond the six hypotheses, we asked whether mixing as such affects polarisation. It does. Final polarisation followed a pronounced inverted U over NormAN proportion (Figure 7). It rose from $M = .026$ in pure-contagion populations to a peak of $M = .097$ at 30% NormAN and then declined to $M = .049$ in pure-NormAN populations. The quadratic trend was strongly negative, $b = -.184$, $t(4397) = -21.38$, $p < .001$ (fitted vertex at 47% NormAN). Mixed populations were much more polarised than pure populations of either type, $M = .077$ vs. $M = .038$, $t(4398) = 19.61$, $p < .001$, $d = 0.77$ (Mann–Whitney U, $p < .001$).

The model contains no negative influence, no homophily, and no biased assimilation, the mechanisms usually needed to produce polarisation in opinion-dynamics models. The disagreement here comes from processing heterogeneity alone. Pure-contagion populations converge by averaging. Pure-NormAN populations converge on a shared pool of evidence. In mixtures, the two update rules pull the population apart. Two cautions apply. First, RMS deviation from the group mean is a coarse dispersion measure, not a measure of clustering into opposing camps [see Hahn et al., 2024]; we use it for continuity with earlier NormAN work. Second, the analysis was exploratory and needs confirmation in an independently designed study.

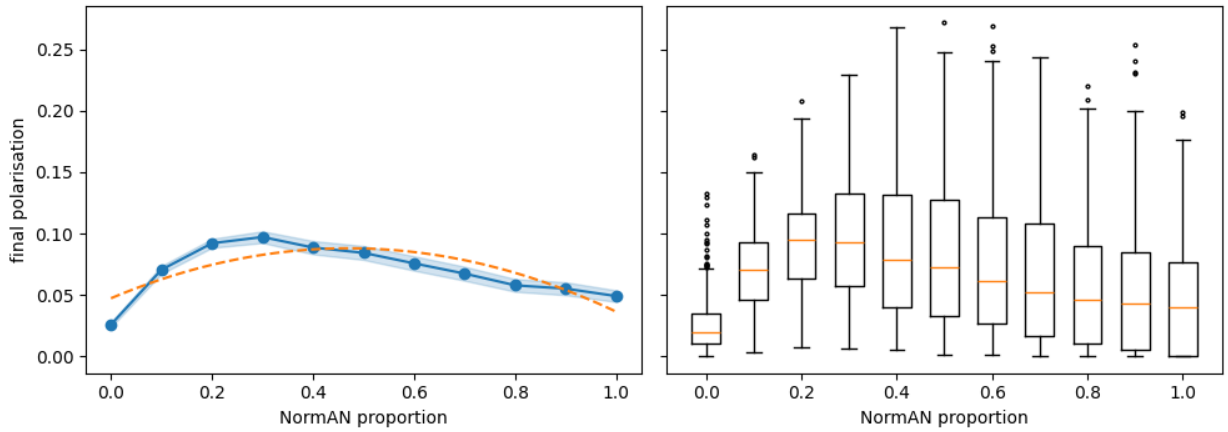


Figure 7: Final polarisation against the proportion of NormAN agents (primary design, 4,400 runs). Left: condition means (points and solid line) with a shaded band of one standard error and a quadratic fit (dashed). Polarisation follows an inverted U, rising from $M = .026$ in pure-contagion populations to a maximum of $M = .097$ near 30% NormAN and falling to $M = .049$ in pure-NormAN populations. Right: run-level distributions at each proportion (box: interquartile range; centre line: median; whiskers: 1.5×IQR; points: outliers), which show the same inverted U in medians and spread. Disagreement is highest in mixed populations, although the model has no negative influence, homophily, or biased assimilation.

4 Discussion

We studied collective belief formation in networks that contain both central-route reasoners, who evaluate the evidential content of arguments, and peripheral-route processors, who track only their neighbours' beliefs. Group accuracy rose with the share of NormAN agents (H1), but non-linearly. Gains were concentrated between 10% and 40% NormAN, which indicates a critical mass rather than a benefit from the first few careful reasoners (H2, in modified form). Impact-based selection

gave no advantage over random sharing and was slightly worse (H3). Broadcast communication sped up consensus and, against our prediction, also improved accuracy (H4a supported, H4b contradicted). Contagion agents at the hubs of a scale-free network harmed accuracy far more than contagion agents at the periphery, an effect about half as large in small-world networks, where hubs barely exist (H5). Contagion agents gained more from mixing than NormAN agents did (H6). And mixing itself produced polarisation, with disagreement highest in mixed populations and lower toward both pure endpoints, in a model without negative influence, homophily, or biased assimilation. We take these findings in turn.

4.1 A population-level formalisation of the ELM

The ELM describes two routes by which one person processes a persuasive message [Petty and Cacioppo, 1986]. Its collective consequences, what happens when a society contains some people who elaborate and many who do not, have not been formalised. Our model is, to our knowledge, the first agent-based model to implement both routes as distinct agent types in one network. In the taxonomy of Assaad et al. [2025], existing models sit at single points on a "reason complexity" axis. Argumentation-based models [Assaad et al., 2023, Olsson, 2013] give every agent propositional reasons; opinion-dynamics models [DeGroot, 1974, Hegselmann and Krause, 2002] give every agent a bare number. Our contribution is not a new point on that axis but a population spread across two of them. The results suggest this is more than a technical variation. Several outcomes, namely the critical-mass threshold, the polarisation peak, and the asymmetric benefit, exist only in mixtures and cannot be seen in any homogeneous model.

The H6 result gives the peripheral route an unexpectedly optimistic reading at the collective level. Peripheral processors are epistemic free-riders in the sense of the literature on the division of cognitive labour [Kitcher, 1990, Weisberg and Muldoon, 2009]. They invest nothing in evaluating evidence, yet their error fell almost fourfold as the share of evidence-evaluating neighbours grew. A society does not need everyone to elaborate for nearly everyone to benefit, because well-founded beliefs spread socially even to those who only track opinions. The caveat matters just as much. The free ride was never complete: a residual gap persisted at every mixing ratio, and the ride exists only if someone else does the epistemic work. This is a population-scale restatement of why the central route matters even when most processing is peripheral.

4.2 Critical mass, not lone voices

H2 predicted that a few careful reasoners would improve accuracy disproportionately, because contagion agents relay their arguments onward. Instead we found a threshold. The first 10% of NormAN agents bought little, and gains accelerated only beyond about a fifth of the population. The forwarding rule suggests why. Contagion agents relay only arguments that match their current lean. When leans are scattered around the uninformed prior, an isolated argument usually dies at its first hop. Only when evidence sources are dense enough that arguments regularly meet already-persuaded relays does diffusion take off.

This pattern converges, through a quite different mechanism, with the experimental finding of Centola et al. [2018] that committed minorities reliably overturn a social convention only once they reach about 25% of the population. It matters for practice. Interventions that seed a few experts, fact-checkers, or high-quality sources into mostly social networks may show near-zero effects, not because the mechanism is wrong but because the seeding is below threshold. Evaluations of such interventions should be designed and powered to detect threshold dynamics, not a linear dose–response.

4.3 Argument reach matters more than argument choice

The two communication factors produced the clearest asymmetry in the study. Optimising argument choice (impact selection) explained less than half a percent of the variance in group error and pointed the wrong way. Argument reach (broadcast vs. pairwise) was the largest factor after population composition (partial $\eta^2 = .136$ vs. .004). In mixed populations, it matters far more that arguments travel than that each one is well chosen.

The H3 null is best read alongside the second case study of Assaad et al. [2023], in which impact-driven sharing undermined consensus in homogeneous NormAN populations. There, senders repeatedly send what they consider their best evidence, less impactful evidence never circulates, and the group settles short of full information. Our results extend that finding to mixed populations and to group error, and suggest that its cause is coverage. Random sharing is a "complete" rule that eventually surfaces every piece of evidence. Best-evidence sharing concentrates traffic on arguments that are redundant from the group's point of view.

Two qualifications apply. First, our impact rule was sender-centric. A recipient-tailored rule, which picks the argument that would most move a particular listener, needs the listener models that Assaad et al. [2023] name as a future direction. It remains untested and might yet rescue the hypothesis. Second, our results do not contradict Schöppl and Hahn [2024], who showed that selection strategy matters in NormAN-only networks. Strategy effects were detectable here too, in the strategy × proportion interaction. They never amounted to the predicted advantage.

The H4 result bears on network epistemology more broadly. In the bandit models of Zollman [2007, 2010], and in the connectivity results of Angere and Olsson [2017] and Hahn et al. [2020], more communication can harm collective accuracy by locking in premature consensus. Here, broadcast communication helped on every measure at every mixing ratio. A plausible reconciliation is that what travels differs. In bandit-style models agents pass on conclusions or noisy signals, which can entrench error. NormAN agents pass on evidence, which keeps its full inferential value whoever relays it. Whether wide, fast communication helps or harms a collective may therefore depend less on bandwidth than on what kind of content is sent: arguments or opinions. This also connects to the flooding-the-zone results of Hahn et al. [2025], where degrading communication (there by dilution, here by restricted scope) slows the diffusion of evidence at a cost to accuracy. The boundary condition should be stated plainly. No premature-consensus penalty emerged within 25 steps. We did not examine longer horizons, over which broadcast-induced consensus might harden before late evidence arrives.

4.4 Structure interacts with composition

The hub-occupation result shows that a network's epistemic performance is not fixed by its composition. Identical populations differed substantially in accuracy depending only on who sat where. A scale-free network can partly make up for a small share of careful reasoners by placing them at its hubs, or waste a large share by ceding those positions to pure amplifiers. The internal check, the effect halving in small-world networks where degree varies little, ties the damage to hub centrality [Barabási and Albert, 1999] rather than to some other property of the seating procedure.

This applies directly to online platforms. There the highest-degree positions are allocated by engagement rather than by quality of elaboration, and a hub occupied by a low-elaboration amplifier is placed to broadcast drift and confirmation-biased forwarding across the network [cf. Cha et al., 2010, on the imperfect link between follower counts and influence]. For intervention design, it implies that relocating or targeting a few structurally central actors may do more than diffuse improvements among many peripheral ones. Extending the hub-occupation manipulation

to targeted "conversion" of individual nodes would test this.

4.5 Polarisation without negative influence

The exploratory finding may carry the most theoretical weight. Classical opinion-dynamics models converge inexorably. Producing sustained disagreement usually requires negative influence, homophily, or biased assimilation [Mäs and Flache, 2013], or bounded-confidence fragmentation [Hegselmann and Krause, 2002]. The NormAN tradition identified another route, incomplete evidence sharing [Assaad et al., 2023, Assaad and Hahn, 2024]. Our model shows a third: heterogeneity in processing. Pure-contagion populations converge by averaging, pure-NormAN populations converge on a shared pool of evidence, and mixtures polarise, with disagreement peaking around a 30% NormAN share. No agent in the model dislikes, distrusts, or avoids anyone. Two connected agents simply update by different rules on the same stream of communication, and that alone holds the population apart.

If real populations mix processing styles, as the ELM claims, some observed polarisation may stem neither from motivated cognition nor from network segregation but from the mundane fact that people update differently. This reframes a normative debate. Interventions that try to reduce polarisation by increasing contact or reducing bias would leave this mechanism untouched. Interventions that shift the distribution of processing styles, or route arguments through evidence-evaluating nodes, would act on it directly. The finding's exploratory status and the coarseness of the dispersion measure [Hahn et al., 2024] both call for replication before this reading is pressed hard. A confirmatory study that varies the mixing ratio with polarisation as the preregistered primary outcome is the obvious next step.

4.6 Limitations

Several limitations bound these conclusions. First, all results come from one world model (the Vole network), one population size ($N = 20$), and one time horizon ($T = 25$). The flooding-the-zone results of Hahn et al. [2025] show that network size can amplify communication effects, so we cannot rule out dependence on scale. Second, the contagion agents' parameters were fixed rather than varied systematically: the susceptibility $\alpha = 0.3$ was set from a preliminary analysis, and the 0.5 threshold that decides which arguments a contagion agent forwards was fixed by design. Full sensitivity analyses remain to be run. Third, the impact rule was sender-centric, which limits what the H3 null says about recipient-tailored argument. Fourth, the binary split between NormAN and contagion agents simplifies what the ELM describes as a continuum of elaboration likelihood moderated by motivation and ability. A graded agent type is the natural refinement. Fifth, the network was static, whereas real ties rewire, partly in response to the very disagreements the model produces. Finally, as with all agent-based models, these results show what mechanisms can produce, not what human populations do. The critical-mass and hub-occupation predictions are, however, concrete enough to test with online experimental collectives of the kind used by Centola et al. [2018].

4.7 Future directions

Four extensions follow directly. The first is recipient modelling. Impact selection over a model of the listener's evidence would test whether argument quality matters once quality is defined from the receiver's side, which completes an agenda item named in the original framework [Assaad et al., 2023]. The second is dynamic networks. Letting ties rewire, by homophily or by preferential attachment to confident speakers, would show whether the polarisation peak hardens into stable echo

chambers or dissolves as mixtures self-segregate. The third is the elaboration continuum. Replacing the binary type with a per-agent elaboration probability would represent the ELM faithfully and would show whether the critical mass is a threshold on people or on elaboration events. The fourth, and most speculative, concerns large language model systems. They now take part in social networks with effectively unbounded connectivity and processing capacity, so they are hubs by construction. The hub-occupation result implies that whether such agents evaluate evidence or amplify engagement will matter out of all proportion to their numbers. The mixed-population framework developed here is a natural testbed for questions about AI participation in public discourse.

5 Conclusion

We formalised, at the level of a population, a theory that has always been about individuals. By implementing the two routes of the Elaboration Likelihood Model as distinct agent types that share one world model and one network, we could ask what happens when a society contains both. Across six hypotheses and one exploratory analysis, the collective outcome was governed by the composition of the population, the reach of its communication, and where its processing types sat in the network, not by the rationality of any individual agent.

Three findings organise the contribution. First, composition dominates, but non-linearly: careful reasoners improve accuracy substantially only once they exceed a critical mass of about a fifth of the population. Second, how far an argument travels matters more than how well it is chosen. Broadcasting evidence widely helped on every measure, while optimising which argument to send did not, because arguments, unlike opinions, keep their inferential value whoever relays them. Third, structure interacts with composition: the same population did better or worse depending only on whether its reasoners or its amplifiers held the hubs. In addition, the exploratory analysis suggests that mixing two processing styles produces sustained polarisation on its own, with no negative influence, homophily, or biased assimilation anywhere in the model.

These results reframe several practical debates. Seeding a few experts into a social network may show near-zero effects simply because the seeding falls below threshold. Reducing polarisation by increasing contact or curbing individual bias would leave the processing-heterogeneity mechanism untouched. And because the damage done by a low-elaboration amplifier scales with its centrality, converting a few highly connected actors may outperform diffuse change among many peripheral ones. This becomes more pressing as large language model systems, hubs by construction, enter public discourse. The conclusions rest on one world model, twenty agents, a fixed horizon, and hand-chosen contagion parameters. They show what mechanisms can produce, and their value lies in the concrete predictions they generate. More broadly, modelling a population as a mixture of processing types, rather than as many copies of one, reveals collective phenomena that no homogeneous model can show.

Code availability

The simulation and analysis code is version-controlled with git and publicly available on GitHub.

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A Assumption checks

Normality of within-cell residuals was assessed for final group error and final polarisation in the primary design ($n = 4{,}400$). Shapiro–Wilk tests rejected normality for both final error ($W = .97$, $p < .001$) and final polarisation ($W = .95$, $p < .001$), with moderate right skew (skewness 0.57 and 0.92). The skew reflects the bounded $[0, 1]$ belief scale, which compresses the lower tail of both measures. At this sample size the Shapiro–Wilk test detects departures far too small to threaten inference, and with 100 replications per cell the sampling distributions of condition means are well approximated as normal. We therefore analysed outcomes parametrically and verified every key pairwise comparison with a Mann–Whitney U test. In every case the non-parametric result agreed with the parametric one.

B ODD protocol summary

The model is described here following the Overview, Design concepts and Details (ODD) protocol [Grimm et al., 2006, 2010, 2020], the standard protocol for describing agent-based models.

Purpose. To investigate how mixing central-route and peripheral-route processing affects collective belief accuracy and polarisation in networked populations.

Entities, state variables, and scales. Two agent types (NormAN and contagion), one social network ($n = 20$), and one shared Bayesian network (Vole, 22 nodes). NormAN agents hold an evidence set $E_i(t)$ and a posterior $b_i(t)$; contagion agents hold a scalar belief $b_i(t)$. Time is discrete, with $T = 25$ steps per run.

Process overview and scheduling. Agents are activated in random order each step, and beliefs are updated synchronously after the communication phase (Algorithm 1).

Design concepts. *Emergence:* collective accuracy and polarisation emerge from individual update rules. *Sensing:* agents observe their neighbours' beliefs and/or arguments. *Interaction:* argument exchange and belief averaging. *Stochasticity:* network generation, initial evidence, activation order, and inquiry are stochastic.

Initialisation. At $t = 0$ a Watts–Strogatz or Barabási–Albert network is generated (Section 2.5) and agents are assigned types according to the condition's NormAN proportion. Each NormAN agent receives one randomly observed evidence node ($k_0 = 1$) and sets its belief by Equation 1. Each contagion agent draws its initial belief from Beta$(2, 2)$.

Input data. None. All evidence is generated from the shared Bayesian network.

Submodels. NormAN belief update (Equation 1), impact selection (Equation 2), contagion belief update (Equation 3), and valence-based forwarding (Equation 4).